

JOAO PESSOA, Brazil

WMO: 827980

Lat: 7.100S

Lon: 34.867W

Elev: 23

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	67.8	68.4	62.8	85.7	81.8	64.3	90.2	79.5	21.6	81.0	20.7	81.1	4.8	210	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	12.6	89.6	76.3	88.2	75.9	87.9	75.8	79.1	85.1	78.5	84.5	78.0	84.1	11.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	142.2	81.6	77.0	140.5	81.3	75.6	134.3	80.5	42.6	85.6	42.0	85.5	41.4	84.5	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.8	16.3	66.6	90.2	1.4	0.9	65.5	90.9	64.7	91.4	63.9	91.9	62.8	92.6		
(5)			64.5	79.8	1.2	1.5	63.7	80.9	63.0	81.7	62.3	82.6	61.5	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.2	80.7	81.0	81.3	80.7	79.5	77.6	76.4	76.2	77.3	78.9	80.1	80.7
	DBStd	2.29	1.42	1.52	1.54	1.61	1.52	1.64	1.38	1.33	1.46	1.35	1.20	1.35
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10655	950	868	970	921	916	829	818	814	820	897	902	950
	CDD65	5180	485	448	505	471	451	379	353	349	370	432	452	485
	CDH74	43875	4705	4478	5090	4346	3618	2451	1922	2083	2560	3582	4266	4774
CDH80	12885	1498	1497	1739	1393	1034	576	333	390	599	1002	1315	1509	
Wind	WSAvg	7.6	7.3	7.2	6.7	6.5	6.9	7.4	8.2	8.6	8.7	8.3	7.8	7.6
Precipitation	PrecAvg	38.2	1.9	2.6	4.1	6.3	6.2	6.8	5.2	2.0	1.3	0.5	0.4	0.9
	PrecMax	60.6	7.4	6.4	8.7	13.1	14.2	13.1	9.9	5.4	4.2	1.8	1.1	3.0
	PrecMin	19.3	0.4	0.2	1.1	1.7	1.1	1.4	1.6	0.4	0.4	0.0	0.0	0.2
	PrecStd	9.9	1.5	1.7	2.2	3.2	3.0	2.8	2.4	1.2	1.0	0.4	0.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	89.9	90.0	89.9	88.2	86.4	84.6	84.6	86.3	87.9	88.2	89.4
		MCWB	75.2	76.3	76.8	76.7	76.8	75.6	73.5	73.1	73.4	73.6	73.8	75.1
	2%	DB	88.1	89.2	89.5	89.2	87.7	85.8	84.2	84.3	85.6	86.3	87.8	88.0
		MCWB	75.3	76.2	76.7	76.8	76.7	75.2	73.6	72.9	73.2	73.5	74.1	74.9
	5%	DB	87.6	87.9	88.1	87.8	86.2	84.4	82.8	82.8	84.3	85.9	86.4	87.6
		MCWB	75.3	76.0	76.5	76.6	76.1	74.5	73.2	72.6	72.8	73.4	73.7	74.8
	10%	DB	86.2	86.4	87.6	86.3	85.7	83.9	82.2	82.4	83.8	84.5	86.0	86.2
		MCWB	75.0	75.9	76.4	76.5	76.0	74.4	73.1	72.5	72.8	73.3	73.6	74.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	79.5	80.0	80.0	79.3	78.0	76.9	76.5	76.5	76.6	77.3	78.3
		MCDB	84.3	85.6	86.0	86.0	85.0	83.3	81.1	81.0	82.0	82.3	84.1	84.1
	2%	WB	77.4	78.4	79.0	79.1	78.3	76.9	75.9	75.3	75.5	75.6	76.3	77.3
		MCDB	83.9	84.7	85.1	85.0	84.2	82.3	80.4	80.4	81.3	82.1	83.4	83.7
	5%	WB	77.0	77.8	78.3	78.3	77.6	76.2	75.1	74.5	74.7	75.2	75.6	76.6
		MCDB	83.7	84.3	84.5	84.2	83.4	81.5	79.9	79.9	80.7	82.0	82.9	83.2
	10%	WB	76.4	77.2	77.7	77.7	77.0	75.5	74.4	73.9	74.2	74.6	75.0	76.0
		MCDB	83.0	83.8	84.0	83.5	82.6	80.8	79.2	79.5	80.2	81.3	82.4	82.7
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.5	12.6	12.3	12.1	12.1	11.9	13.2	13.2	13.1	13.2	12.7
		MCDBR	13.5	13.5	13.5	13.8	13.3	13.4	13.2	14.0	14.0	13.9	13.7	13.5
		MCWBR	4.3	4.5	4.6	4.9	5.3	5.8	5.8	5.9	5.1	4.6	4.4	4.2
	5% WB	MCDBR	12.2	12.3	12.4	12.2	12.1	12.0	11.4	12.7	12.9	13.0	13.1	12.4
		MCWBR	4.6	4.9	5.0	5.2	5.4	5.9	5.8	6.3	5.5	4.8	4.8	4.5
Clear-Sky Solar Irradiance	taub	0.397	0.401	0.400	0.397	0.388	0.390	0.393	0.403	0.424	0.417	0.403	0.411	
	taud	2.494	2.491	2.499	2.532	2.553	2.538	2.512	2.446	2.365	2.388	2.439	2.433	
	Ebn at Noon	299	297	292	285	279	273	275	279	281	288	295	293	
	Edh at Noon	36	37	36	33	31	31	32	36	40	40	38	38	
All-Sky Solar Radiation	RadAvg	1940	1934	1914	1719	1571	1418	1479	1715	1877	2010	2089	2004	
	RadStd	116	93	96	113	110	87	105	72	86	76	69	72	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	-0.85	-0.54	-1.17	-1.71	-0.83	-0.59	N/A	N/A	-392	-392

Nomenclature: See separate page